

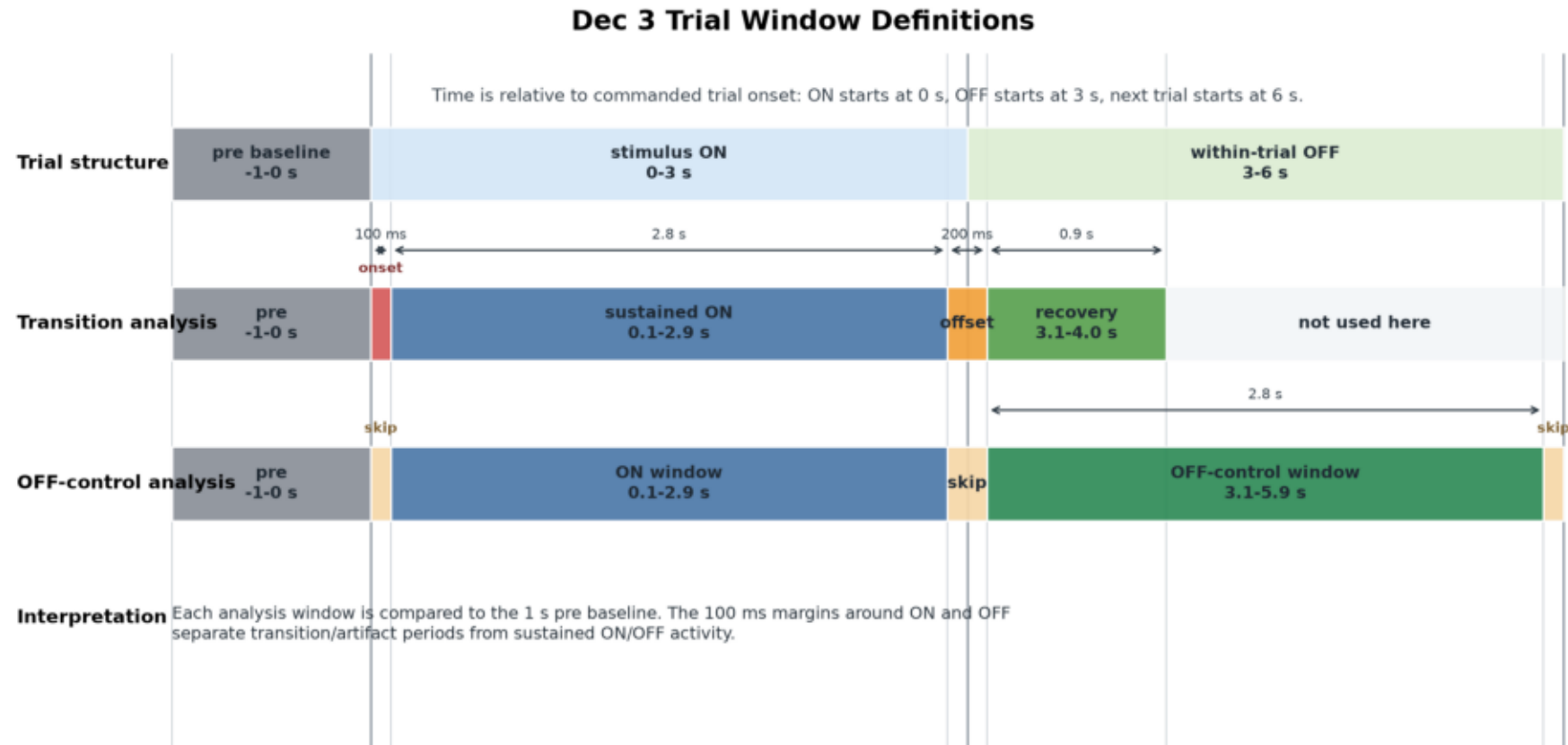
Can we drive the brain through the skin?

Single-unit and LFP responses to amplitude- and frequency-varied haptic stimulation in hippocampus & entorhinal cortex

Motivation: peripheral routes to non-invasive neuromodulation

Before targeting brain states via the skin, we need to know whether — and how — a cutaneous stimulus is represented centrally.

Design: parametric haptic stimulation with simultaneous electrophysiology



3 s ON / 3 s OFF blocks across four carrier frequencies (5/10/26/50 Hz) and three amplitudes; ~200 repeats/condition; dual-region linear-probe recording.

Three questions, increasing in strength of claim

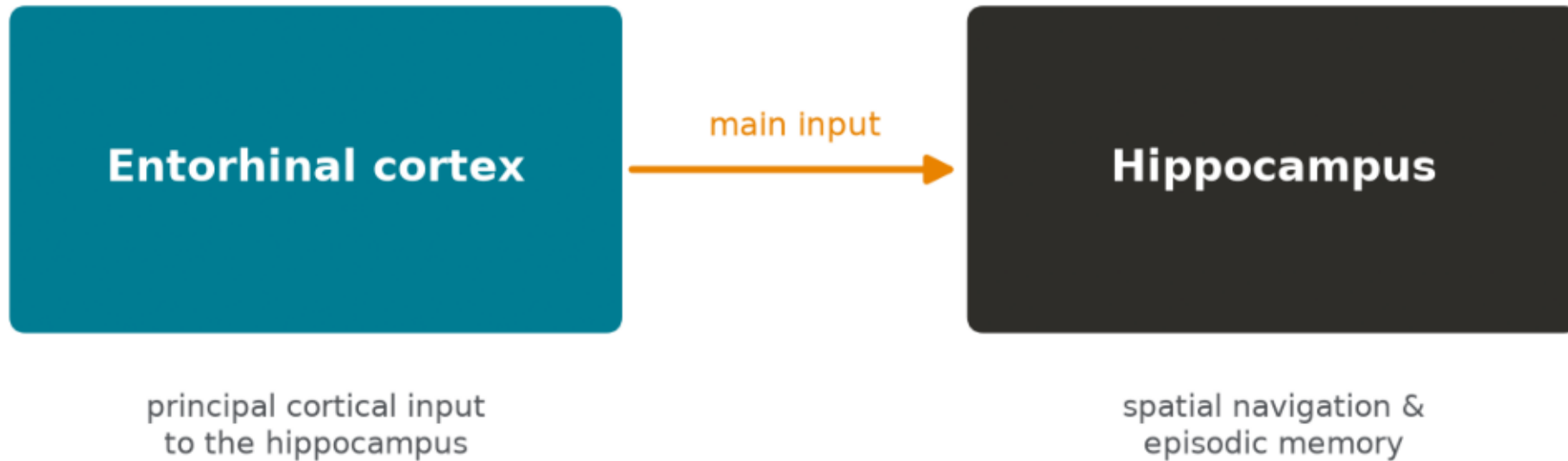
(1) Is there a central response? (2) Is it frequency-specific? (3) Does it entrain neural activity (phase-locking)?

Part 1 — Two readouts, two regions

The minimum background, with the key methodological distinction

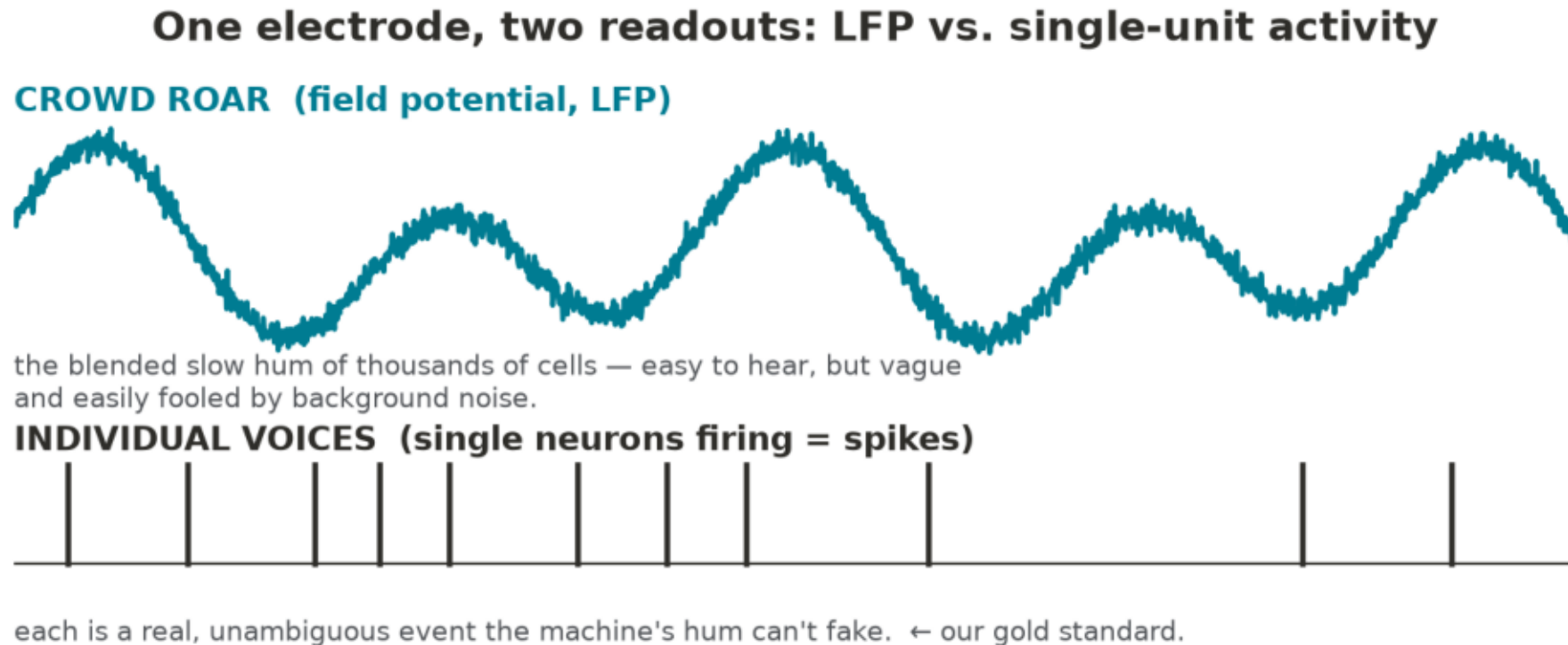
Recording sites: hippocampus and entorhinal cortex

Two connected regions, recorded simultaneously



Entorhinal cortex = principal cortical input to hippocampus; hippocampus = spatial/mnemonic processing. A connected circuit, recorded simultaneously.

One electrode yields two readouts: LFP and single-unit activity



LFP = low-frequency aggregate field (synaptic/transmembrane currents of many cells). Spikes = action potentials of individual, sorted neurons. They differ in what can fake them.

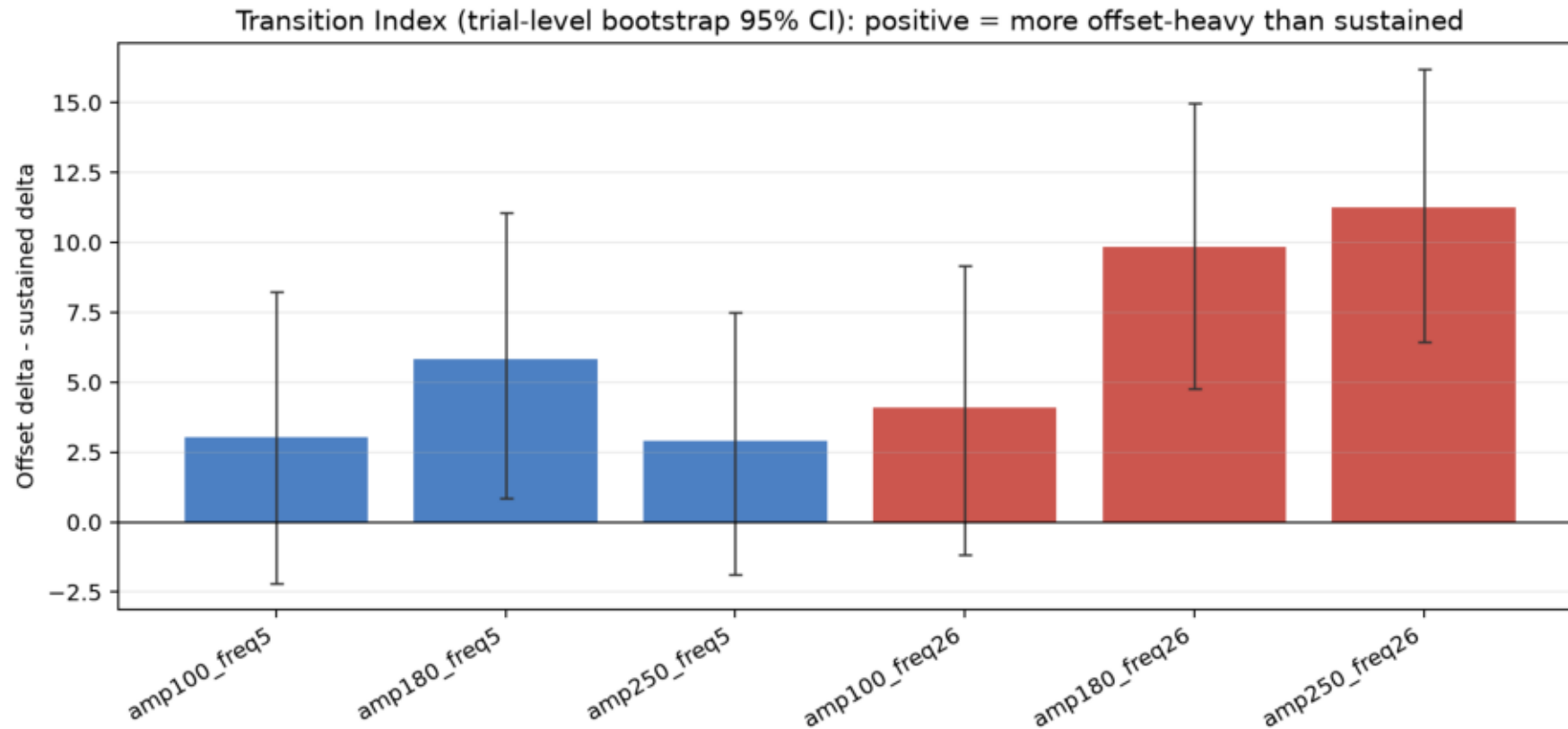
Single units are the more conservative readout

A sorted action potential is a discrete biological event; the LFP can be mimicked by volume-conducted electrical artifact. Spikes constrain interpretation far more tightly.

Part 2 — Results

FINDING 1

A robust LFP response — transition-weighted, not a sustained oscillation

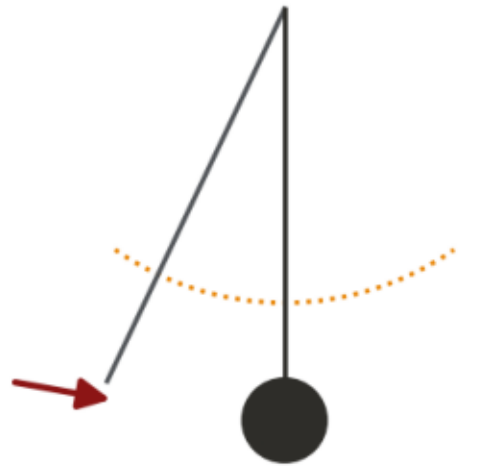


The broadband LFP responds at stimulus onset/offset (an evoked transient), not as a sustained rhythm at the carrier frequency.

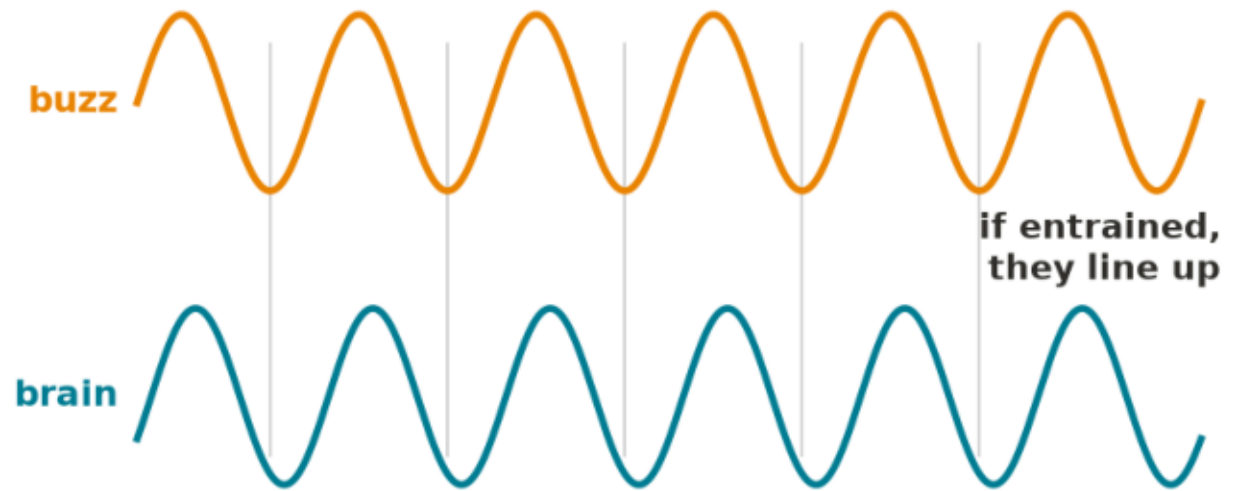
DEFINITION

Entrainment, defined: phase-locking of neural activity to the stimulus

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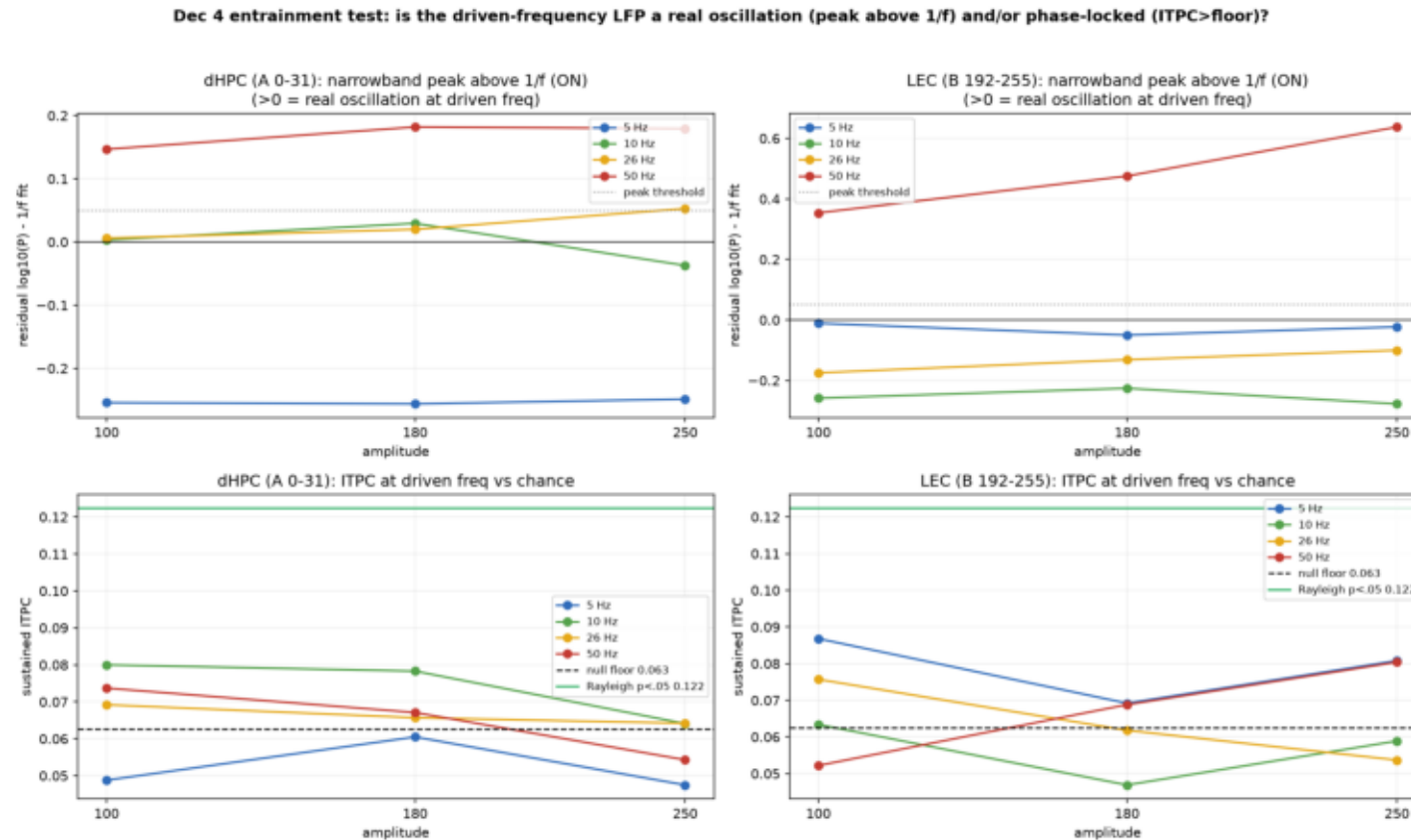
push in rhythm →
the swing goes higher



Entrainment = endogenous activity aligns to the stimulus phase (a driven-oscillator relationship). The strong claim — and the most useful, if true.

FINDING 2

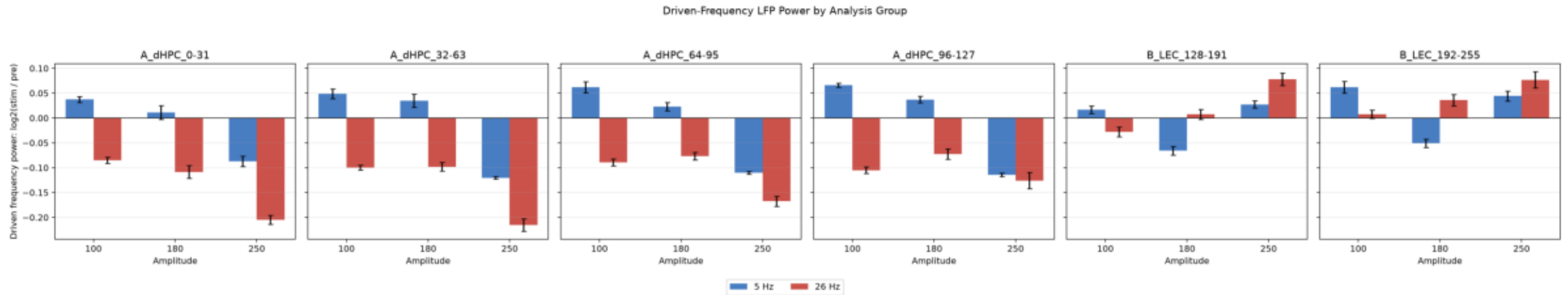
No frequency-following in hippocampus at any carrier



No narrowband spectral peak above the $1/f$ background, and inter-trial phase consistency at chance — across 5/10/26/50 Hz, replicated across sessions.

FINDING 3

Entorhinal cortex shows an amplitude-graded 50 Hz LFP increase



A narrowband 50 Hz LFP power increase, scaling with amplitude — the candidate entrainment signal. But the LFP is exactly the readout most vulnerable to artifact.

The confound: stimulator-coupled 50 Hz electrical artifact

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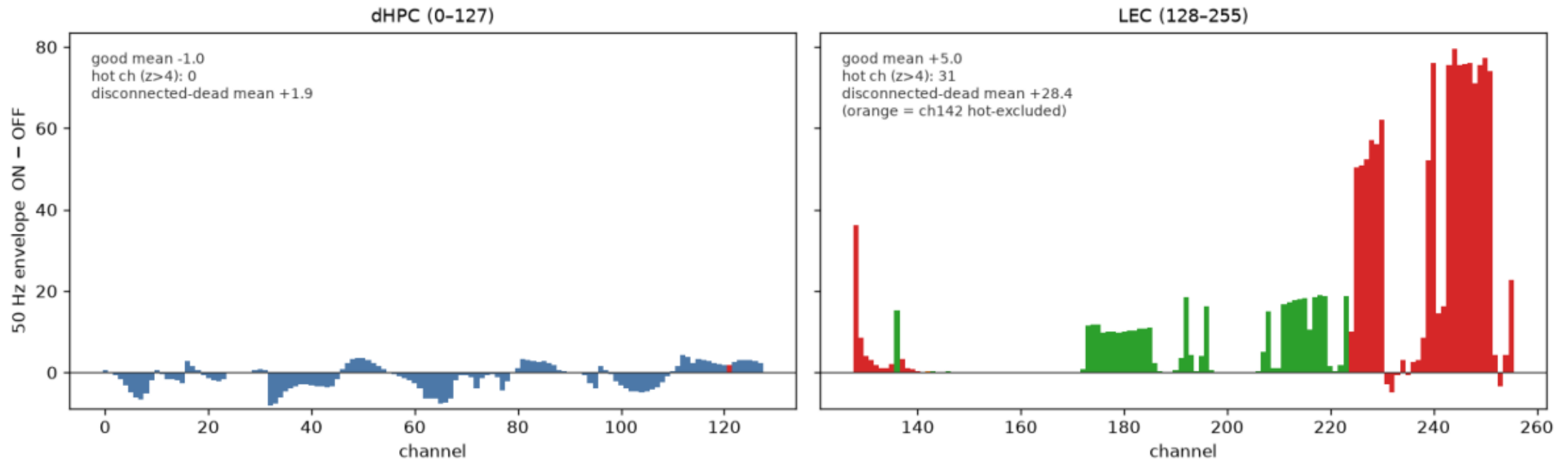
The mic records BOTH → a clean '50 Hz' could be the SPEAKER, not the brain.

An electromechanical actuator can inject a 50 Hz electrical artifact directly into the recording — indistinguishable from a neural 50 Hz in the LFP alone.

FINDING 3, CONTROLLED

Disconnected channels carry the 50 Hz — so it is largely pickup

Is the 50 Hz pickup gradient LEC-only? dHPC (left): flat, ~ 0 , no hot channels, dead ch clean. LEC (right): rises toward the dead blocks (128-135, 224-255).

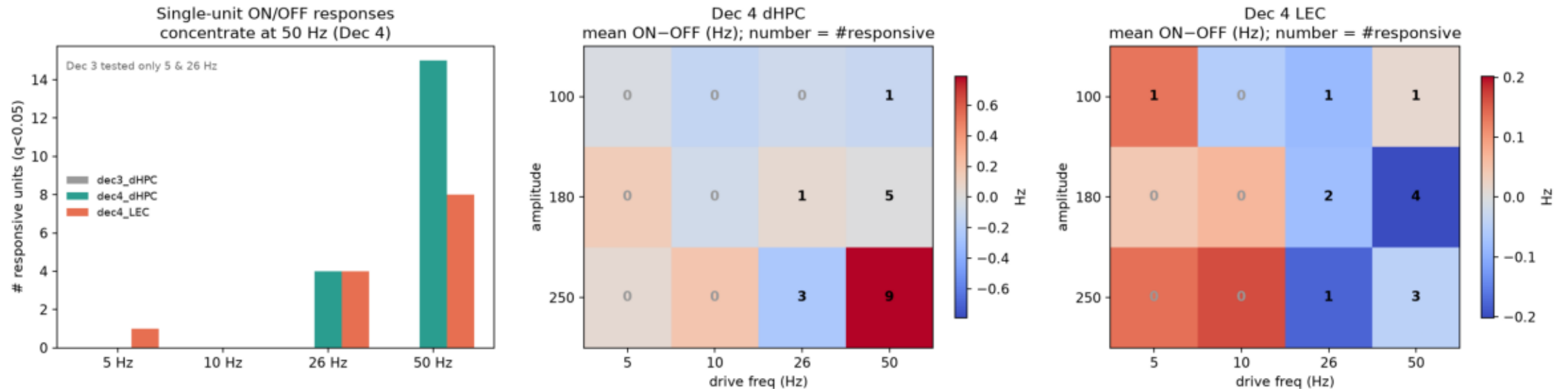


QC-failed/disconnected electrodes (which cannot record neural signal) show $\sim 6\times$ more 50 Hz than in-tissue channels. The entorhinal 50 Hz LFP is substantially artifact.

FINDING 4 — THE ROBUST RESULT

Frequency-specific single-unit rate modulation at 50 Hz

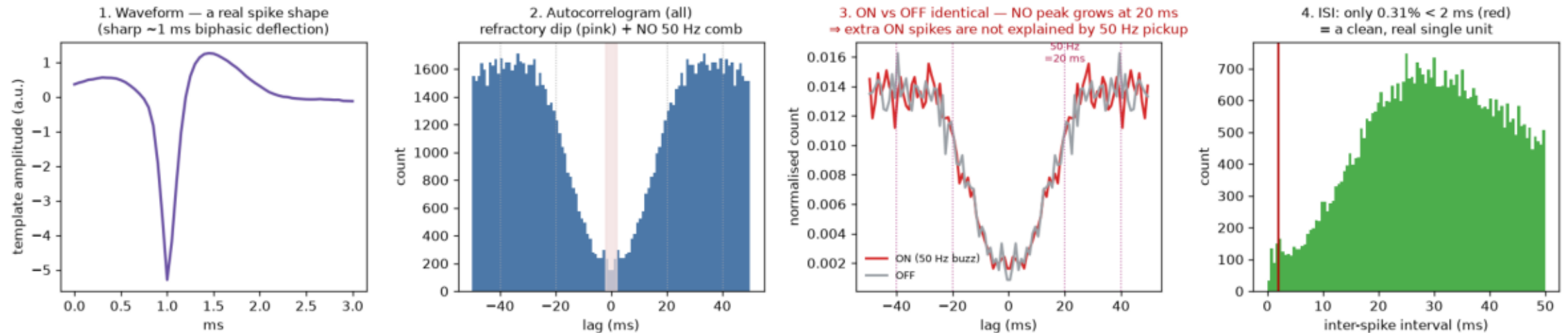
Curated single-unit ON-vs-OFF firing — Dec 3 null at 5/26 Hz; Dec 4 effect at 50 Hz / high amplitude



Sorted single units change firing rate during 50 Hz / high-amplitude stimulation, in both regions — an effect electrical pickup cannot produce.

Controls: high-pass separation + autocorrelogram / ISI screens

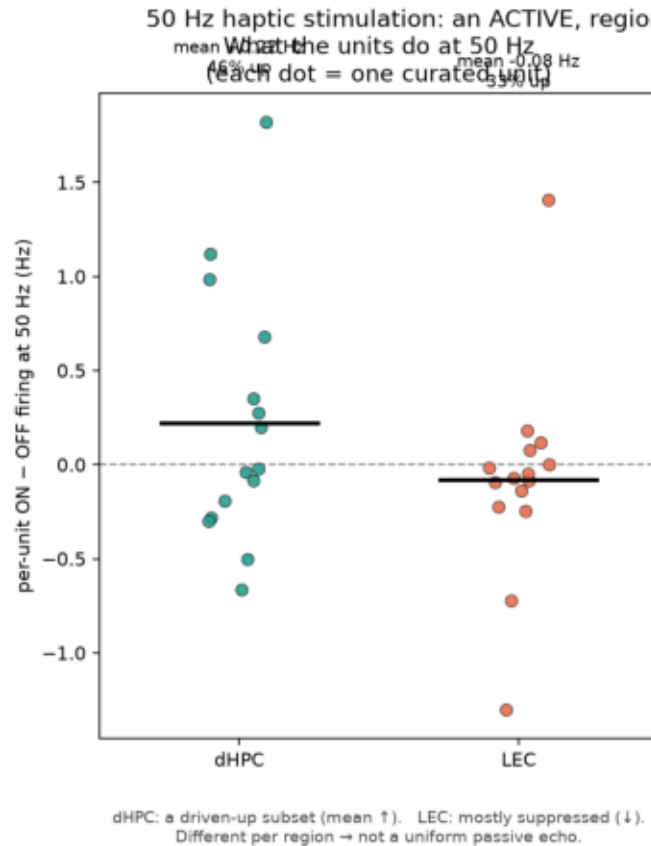
UNIT 87 — the 'Phy view', computed: the soft-spot LEC unit is a REAL neuron, and its 50 Hz ON rate increase is not explained by 50 Hz pickup (its autocorrelogram grows no 50 Hz periodicity during ON).
Comprehensive screen: 0/8 up-going units show an ON 50 Hz comb, and 0/8 show an ON ISI<2ms rise $\Rightarrow$ the up-going rate increases are not explained by 50 Hz pickup.



Spikes are detected $> \sim 300$ Hz (50 Hz pickup is removed pre-detection); units show no stimulus-locked 50 Hz periodicity and no ON-rise in refractory violations.

FINDING 5

Region-specific, bidirectional modulation — not a passive relay



What does the 50 Hz response mean? — the inference ladder

X Is it electrical / mechanical ARTIFACT?

NO — sorted single units change their firing RATE; pickup can't do that.

X Is it a PASSIVE ECHO (brain as a relay)?

UNLIKELY — regions respond DIFFERENTLY (dHPC driven-up subset; LEC suppressed) and the LFP 50 Hz is induced, not phase-locked. A relay would look the same everywhere.

✓ ACTIVE, region-specific processing?

SUPPORTED — the 50 Hz input is transformed differently by each circuit (not relayed).

? Are the regions COORDINATING ("working together")?

NOT TESTED — needs spike-field coupling + cross-region coherence. We showed each region responds, not that they coordinate.

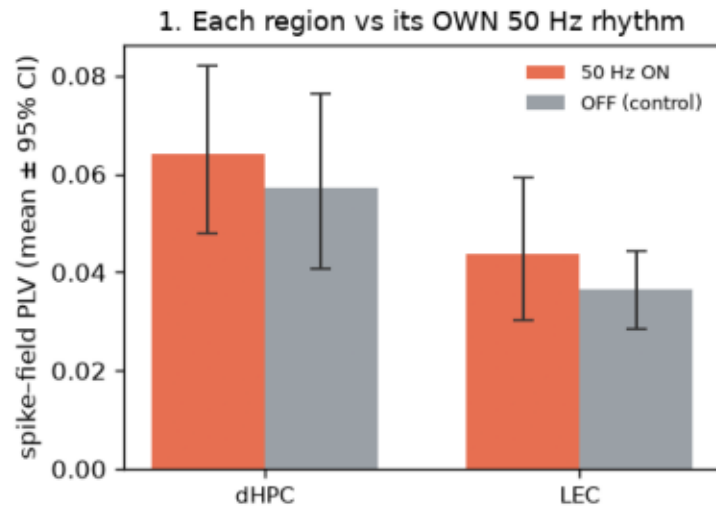
Hippocampus: a driven-up subset. Entorhinal cortex: net-suppressed. Opposite transformations of identical input argue for active, circuit-specific processing.

FINDING 6

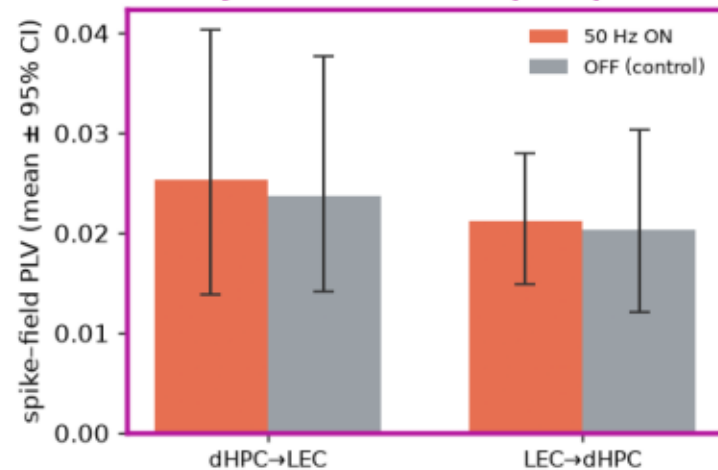
No clear cross-regional coordination at 50 Hz

Do dHPC & LEC 'work together' at 50 Hz? — If they did, panel 2 (cross-region SPIKES) would rise during ON; it does NOT. Panel 3 (LFP) rises but is fooled by a shared signal. $\Rightarrow$ NO clear coordination.

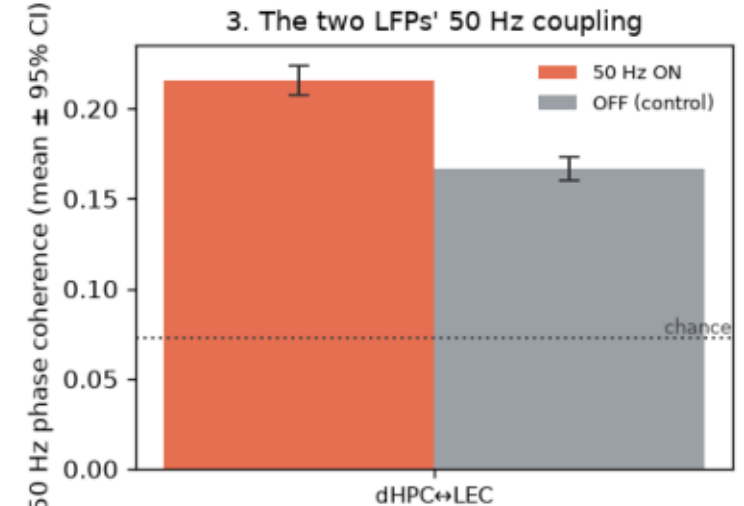
2. SPIKES vs the OTHER region's rhythm (the key test — harder for pickup to fake)



Weak; ON $\approx$ OFF (CIs overlap).
Baseline gamma coupling,
not stimulus-driven.



Very weak; ON $\approx$ OFF (CIs overlap).
Neurons do NOT track the other
region $\rightarrow$ no coordination.



ON > OFF (CIs separate) — looks like
coupling, BUT inflated by a SHARED
signal; spikes (panel 2) don't follow
it $\rightarrow$ not real coordination.

50 Hz trials, all amplitudes pooled (amp100+180+250, 600 trials). ON = 3 s buzz; OFF = following 3 s gap.

LFP-LFP coherence rises, but the artifact-resistant cross-region spike-field measure does not — parsimoniously a shared signal, not interaction.

Part 3 — The entrainment question, and an instrumentation limit

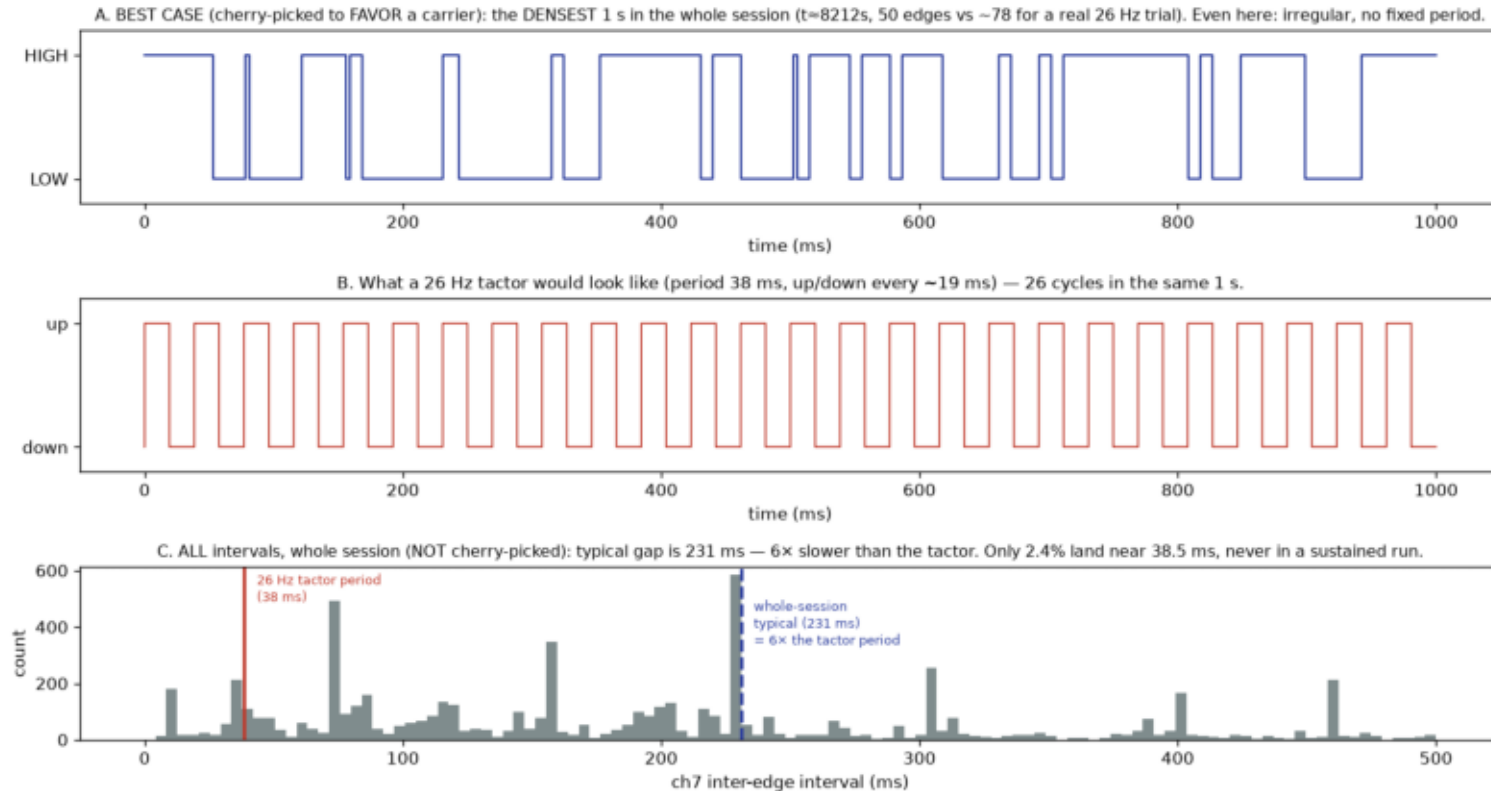
Entrainment was untestable — a measurement gap, not a null

Phase-locking to the stimulus requires a recorded, time-aligned stimulus-phase reference. We lacked one — so entrainment is untestable here, not negative.

DIAGNOSIS

The digital sync channel never captured the carrier

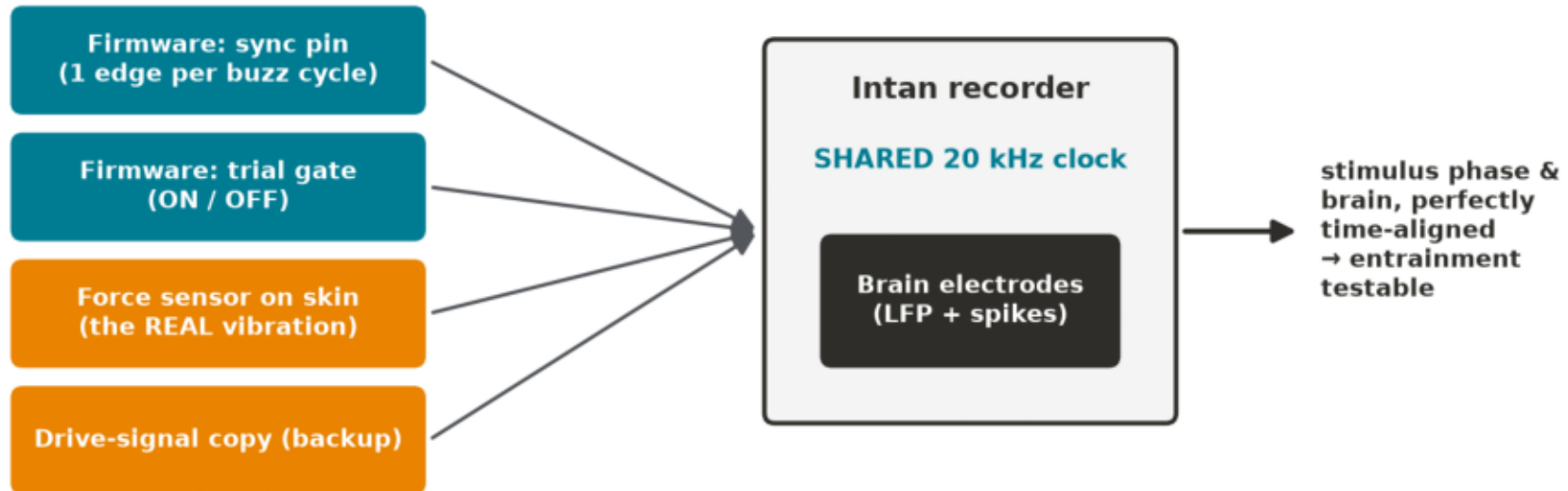
ise. Panel A is the BEST/densest window (cherry-picked to favor a carrier); Panel C is ALL data (typical). ch7 toggles $\sim 4\times/s$; a 26 Hz tactor toggles $26\times/s \rightarrow$ tactor does ~ 6 full



The recorded sync line updated at ~ 4 Hz, independent of carrier (identical pulse counts at 5 vs 26 Hz; ~ 78 expected at 26 Hz). No phase reference exists in the data.

Fix: redundant, clock-shared stimulus references

Fix: record the stimulus 3 ways on the acquisition clock



+ a 2-minute verification recording BEFORE the real session

Firmware per-cycle sync + a transduced force/accelerometer signal + a drive-signal copy, all on the acquisition clock — cross-checked, with pre-session verification.

Summary

Genuine central response to haptic stimulation; cleanest evidence is frequency-specific, region-specific single-unit modulation at 50 Hz; the 50 Hz LFP is largely artifact; entrainment untestable here.

Implications & next step

Establishes a measurable central readout of peripheral haptic input and the controls to interpret it; with the instrumentation fix, entrainment becomes directly testable.

Thank you — questions

Happy to go into any figure, the artifact controls, or the next-round instrumentation.